

NeckarCut

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Abstract

The Maximum Agreement Forest (MAF) problem is a fundamental challenge in computational phylogenetics, seeking the minimum number of edge cuts required to decompose a set of trees into a common subforest. In this paper, we present an exact Branch-and-Bound solver developed for the Exact Track of the PACE 2026 challenge. Our algorithm generalizes classic 3-way branching rules for structural conflict resolution to operate over an arbitrary number of input trees.

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1 Introduction

The Maximum Agreement Forest (MAF) problem asks for the minimum number of edge cuts required to decompose a set of phylogenetic trees into a common forest. This problem is of central importance in computational biology for understanding reticulate evolution and modeling phylogenetic networks. As the problem is known to be NP-hard, even for two trees, exact solutions require careful exploration of the search space.

In this work, we present an exact Branch-and-Bound solver designed to compute the MAF for a given set of input trees. By systematically identifying conflicting sibling pairs and branching on the possible ways to resolve these conflicts, we guarantee an optimal solution.

2 Preliminaries

Let $\mathcal{T} = \{T_1, T_2, \dots, T_t\}$ be a list of phylogenetic trees on the same leaf-set X , where $n = |X|$ is the number of leaves. An *agreement forest* for $\mathcal{T}$ is a forest F of phylogenetic trees whose leaf-set is X such that F can be obtained from each $T_i \in \mathcal{T}$ by first removing directed edges and then exhaustively contracting vertices that are not in X but have in- and out-degree at most one. The Maximum Agreement Forest problem seeks an agreement forest with the minimum number of components k . The number of components is equal to the number of edge cuts plus one.

In our algorithm, a pair of leaves (a, b) are *siblings* if they share the same parent node. A leaf that is completely disconnected from the rest of the tree is referred to as a *single node*.

3 Related Work

The computation of Maximum Agreement Forests is a classic problem in fixed-parameter tractability (FPT). For two trees, early FPT algorithms achieved running times of $O(3^k n)$, where k is the number of edge cuts (or SPR distance). This is largely based on the observation that for any conflicting sibling pair, one must cut the edge above one of the siblings or the path connecting them. This foundational 3-way branching strategy to resolve sibling conflicts was detailed in exact branch-and-bound algorithms by Whidden, Beiko, and Zeh [2].

Extensions of the MAF problem to multiple trees ($t > 2$) have been studied by Shi et al. [1], showing that the problem can be solved in $O(3^{kt} n)$ time or $O^*(2.42^k n^4 t^3)$ time. Our

42 solver leverages the standard 3-way branching rules [2] and generalizes them to operate
 43 efficiently over an arbitrary number t of input trees.

44 **4 Exact Algorithm**

45 Our solver employs a recursive Branch-and-Bound strategy to find the minimum number of
 46 edge cuts necessary to reach an agreement forest.

47 **4.1 Core Algorithm**

48 The core algorithm iteratively computes the Maximum Agreement Forest by comparing a
 49 running reference forest F_0 against the input trees sequentially. Initially, the reference forest
 50 is set to the first tree in the sequence, that is $F_0 = \{T_1\}$. We then compare F_0 against the
 51 next tree in the sequence, which is contained in its own forest $F_i = \{T_i\}$. Once all conflicts
 52 between F_0 and F_i are resolved, the resulting intermediate agreement forest becomes the new
 53 F_0 , and the algorithm proceeds to the next forest $F_{i+1} = \{T_{i+1}\}$, continuing this process
 54 until all t trees have been reconciled.

55 To resolve structural differences between F_0 and the current target F_i , the algorithm
 56 locates a sibling pair (a, b) in F_i and examines their topological relationship in F_0 :

- 57 **1. Siblings in F_0 :** If a and b are also siblings in F_0 , no conflict exists for this pair between
 58 the two forests. We safely apply a reduction rule by shrinking a and b into a new combined
 59 leaf in both forests.
- 60 **2. Disconnected in F_0 :** If a and b reside in different connected components in F_0 , they
 61 have already been separated by a previous cut. They cannot be siblings in the final
 62 agreement forest. We resolve this by branching into two subproblems:
 63 – Cut the edge connecting a to its parent.
 64 – Cut the edge connecting b to its parent.
- 65 **3. Connected but not siblings in F_0 :** If a and b are in the same component in F_0 but
 66 separated by a path of internal nodes, there is a structural conflict. We branch into three
 67 subproblems to resolve this:
 68 – Cut the edge connecting a to its parent.
 69 – Cut the edge connecting b to its parent.
 70 – Cut all edges in F_0 such that (a, b) becomes a sibling.

71 **4.2 Preprocessing**

72 Before initiating the branch-and-bound search, we parse the input and greedily scan all trees
 73 for pairs of leaves that are siblings across *all* input trees. These universally agreeing sibling
 74 pairs are permanently shrunk into a single new leaf. This safe reduction rule shrinks the
 75 problem size without compromising optimality.

76 **4.3 Tree Ordering Heuristic**

77 The order in which trees are processed impacts the ability to find a tight upper bound early.
 78 We compute a structural distance estimate $D(F_1, F_2)$ between two forests F_1 and F_2 based
 79 on the mismatch of sibling pairs. Let $S(F)$ be the set of sibling pairs in a forest F , and let
 80 $d_F(a, b)$ denote the path distance between leaves a and b in forest F (measured as the number

of edges between them, or the sum of their depths if they are in different components). The distance is defined as:

$$D(F_1, F_2) = \sum_{(a,b) \in S(F_1)} d_{F_2}(a, b) + \sum_{(c,d) \in S(F_2)} d_{F_1}(c, d) \quad (1)$$

The algorithm initially orders the forests such that the two most structurally dissimilar trees (maximizing D) are evaluated first. Dynamically during the search, before reconciling with the next tree, the solver reorders the remaining candidate forests to choose the one that maximizes the distance $D(F_0, F_{\text{candidate}})$ relative to the current reference forest F_0 . This rapidly increases the number of cuts in the initial branches, yielding a strong bound that allows for aggressive pruning later.

4.4 Bounding and Pruning

We maintain a global upper bound on the minimum number of trees (components) found so far, initially set to the trivial upper bound. At any state in the search tree, if the number of current components in F_0 plus the number of separated single nodes meets or exceeds the current best minimum, the branch is immediately pruned.

4.5 Sketch of Correctness

The correctness of our exact Branch-and-Bound solver relies on the exhaustiveness of the 3-way branching rule. When reconciling F_0 and F_i , any structural conflict identified by a sibling pair (a, b) in F_i implies that at least one cut must be made to form an agreement forest. The three branches considered—cutting the edge above a , cutting the edge above b , or cutting an edge on the path connecting a and b in F_0 —exhaustively cover all possible ways to disconnect the topological relationship that contradicts (a, b) being siblings. Because the search tree explores all such minimal resolutions and bounds them against the global optimal size found so far, it is guaranteed to explore the search space completely without missing the optimal agreement forest.

References

- 1 Feng Shi, Jianer Chen, Qilong Feng, and Jianxin Wang. A parameterized algorithm for the maximum agreement forest problem on multiple rooted multifurcating trees. *Journal of Computer and System Sciences*, 97:116–129, 2018.
- 2 Chris Whidden, Robert G. Beiko, and Norbert Zeh. Fast FPT algorithms for computing rooted agreement forests: Theory and experiments. In Paola Festa, editor, *Experimental Algorithms, 9th International Symposium, SEA 2010, Ischia Island, Naples, Italy, May 20-22, 2010. Proceedings*, volume 6049 of *Lecture Notes in Computer Science*, pages 141–153. Springer, 2010. doi:10.1007/978-3-642-13193-6_13.